

CROWN & PITCH LLC

Payment Checkout — Developer Implementation Brief

Bank of America Secure Acceptance | Registration Checkout Flow

Version 1.0 | March 18, 2026

1. Purpose & Scope

This brief defines all requirements for implementing the payment checkout updates on the existing Crown & Pitch LLC live website. The site is already live. This work adds the payment method selection flow, fee and tax calculation logic, required disclosures, and Bank of America (BoA) Secure Acceptance API integration to the existing registration system.

The developer must implement all items in this brief before activating the payment feature. The existing site must remain fully operational throughout — all changes should be built and tested on a staging branch before being pushed to production. Nothing in this brief is optional unless explicitly labeled as such.

2. Payment Gateway — Bank of America Secure Acceptance

2.1 Integration Method

Use the Bank of America Secure Acceptance Checkout API (v1.1.0). This is a server-side signed POST implementation — payment card data flows directly from the customer's browser to BoA servers and never touches Crown & Pitch servers. Integration should be added to the existing Next.js site without disrupting any currently live pages or functionality.

Integration Type	Checkout API (Secure Acceptance)
Test Endpoint	https://testsecureacceptance.merchantservices.bankofamerica.com/silent/pay
Production Endpoint	https://secureacceptance.merchantservices.bankofamerica.com/silent/pay
Transaction Types	sale sale,create_payment_token
Supported Cards	Visa (001), Mastercard (002), American Express (003), Discover (004)
Currency	USD only
Auth Method	HMAC-SHA256 signed fields

2.2 ACH / Bank Transfer

ACH is NOT available through the BoA Secure Acceptance integration at this time. The payment_method field supports card only. BoA is investigating whether their ISV can provide ACH separately — this is a future enhancement. Do not hold up or block the current checkout implementation waiting for ACH resolution.

2.3 Required Signed Fields

All of the following fields must be included in the `signed_field_names` parameter and signed with HMAC-SHA256 before submission. Sign everything except `card_number`, `card_cvn`, and `signature`.

```
access_key
amount
currency
locale
payment_method
profile_id
reference_number
signed_date_time
signed_field_names
tax_amount
transaction_type
transaction_uuid
unsigned_field_names
```

⚠ The `signed_date_time` field must be accurate UTC time. Stale timestamps will cause transaction failures.

3. Payment Method Selection — Split Button Flow

The BoA gateway does not automatically detect or distinguish between credit and debit cards. Crown & Pitch cannot apply a processing fee to debit card transactions (prohibited under Texas law and federal Durbin Amendment). The solution is a customer-selected payment path shown before card entry.

3.1 Button Display

Before showing any card entry form, display two clearly labeled payment path buttons side by side:

Pay with Credit Card 3% processing fee applies	Pay with Debit Card No processing fee
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3.2 Behavior by Path

Credit Card path: Show full fee breakdown including processing fee. Calculate final amount = registration fee + tax + processing fee. Pass this total as the amount field to BoA.

Debit Card path: Show fee breakdown without processing fee line. Calculate final amount = registration fee + tax only. Pass this total as the amount field to BoA.

⚠ There is no technical enforcement of card type. A customer could select Debit and enter a credit card number. This is accepted industry practice — the customer's self-selection plus the disclosure language provides legal coverage. Do not attempt to build card-type detection; it is not supported by the BoA gateway.

4. Fee & Tax Calculation Logic

4.1 Rates

Texas Sales Tax	8.25% (6.25% state + 2.0% Prosper local)
Processing Fee	3.00% — credit card path only
Processing Fee — Debit	0% — never applied to debit path

4.2 Calculation Order

Calculate in this exact sequence. Tax is calculated on the registration fee only. The processing fee is calculated on the combined total of registration fee plus tax.

```
// CREDIT CARD PATH
registration_fee = [base fee for selected tournament/division]
tax_amount      = registration_fee × 0.0825
subtotal        = registration_fee + tax_amount
processing_fee   = subtotal × 0.03
total_due       = subtotal + processing_fee

// DEBIT CARD PATH
registration_fee = [base fee for selected tournament/division]
tax_amount      = registration_fee × 0.0825
total_due       = registration_fee + tax_amount
// No processing fee line
```

4.3 BoA API Amount Fields

Two specific fields must be sent to the BoA API on every transaction:

amount	The full total_due value — this is the amount BoA will charge the card
tax_amount	The tax_amount value only — sent separately so BoA can report it correctly

⚠ The amount field must equal total_due including all fees. Do not send the pre-fee subtotal and expect BoA to add fees — BoA charges exactly what is in the amount field.

4.4 Rounding

Round all calculated values to 2 decimal places using standard rounding (half-up) before display and before passing to the BoA API. Never truncate.

5. Checkout Screen Layout & Display Requirements

5.1 Fee Breakdown — Credit Card Path

Display all line items in this exact order above the payment button. All four lines are mandatory on the credit card path:

Line Item	Amount
Registration Fee	\$XXX.00
Texas Sales Tax (8.25%)	\$XX.XX
Processing Fee (3% — Credit Card)	\$X.XX
TOTAL DUE	\$XXX.XX

5.2 Fee Breakdown — Debit Card Path

Display three lines only. The processing fee row must not appear on the debit path:

Line Item	Amount
Registration Fee	\$XXX.00
Texas Sales Tax (8.25%)	\$XX.XX
TOTAL DUE	\$XXX.XX

5.3 Display Rules

- All amounts must display as dollar values with exactly 2 decimal places (e.g., \$12.50, not \$12.5)
- The fee breakdown table must appear above the Pay button — never after
- The processing fee line must show both the percentage label AND the dollar amount
- Both the percentage rate and dollar amount must be dynamically calculated — no hardcoded display values
- Page-level notice (above the registration form, before the customer starts filling out): "A 3% processing fee applies to credit card payments. Choose Pay with Debit Card to avoid this fee."

6. Required Disclosure Text

The following disclosure language is legally required and must appear verbatim at the locations specified. Do not paraphrase or abbreviate.

6.1 Page-Level Notice

Location: Top of the registration/pricing page, above the registration form, before any form fields.

A 3% processing fee applies to credit card payments. Choose "Pay with Debit Card" to avoid this fee.

6.2 Checkout-Level Disclosure

Location: Directly above the Pay button on the credit card path. Must be visible before the customer submits payment.

A 3% processing fee is applied to credit card transactions to offset payment processing costs. This fee is non-refundable. Select “Pay with Debit Card” above to avoid this fee.

This disclosure must appear on the credit card path only. It must not appear on the debit card path.

6.3 Formatting Requirements for Disclosures

- Font size: minimum 12pt (20 half-points in the API) — must not be smaller than surrounding body text
- Color: must not be grayed out, light, or visually de-emphasized
- Must not be hidden behind a toggle, tooltip, accordion, or modal
- Must be fully visible without scrolling on standard desktop viewport (1280px wide)

7. Receipt & Confirmation Requirements

7.1 On-Screen Confirmation Page

After BoA returns a successful transaction response (decision = ACCEPT, reason_code = 100), display a confirmation page that includes all of the following:

- Tournament name and division registered for
- Team/player name
- Registration Fee — dollar amount
- Texas Sales Tax (8.25%) — dollar amount
- Processing Fee (3%) — dollar amount (credit card path only; omit on debit path)
- Total Charged — dollar amount
- Last 4 digits of card used (from req_card_number response field — BoA returns this masked)
- BoA Transaction ID (from transaction_id response field)
- Date and time of transaction

7.2 Email Receipt

BoA must be configured to send an automatic email receipt to the customer. Configure this in the Secure Acceptance profile under Notifications > Email Receipt to Customer. The receipt email must also include all line items listed in Section 7.1.

Sender Email	To be configured — use the Crown & Pitch noreply address
Sender Name	Crown & Pitch LLC
Send Copy To	Crown & Pitch admin email address

7.3 Merchant POST URL

Configure the Merchant POST URL in the Secure Acceptance profile. BoA will send a full transaction result payload to this URL for every transaction, including declined ones. This is the backup confirmation mechanism if the customer's browser disconnects after payment.

- URL must be HTTPS with TLS 1.2 or higher

- Server must respond within 10 seconds or BoA may delay future notifications
- Parse and store transaction result data to the order management system on every POST

7.4 Declined Transactions

If BoA returns decision = DECLINE or decision = ERROR, display a clear error message to the customer and allow them to re-enter payment details or return to payment method selection. Do not treat a failed attempt as a completed registration.

8. Security & PCI Compliance

8.1 Card Data Handling

Crown & Pitch servers must never receive, store, log, or process raw card numbers, CVN codes, or expiration dates. The Secure Acceptance Checkout API is architected to prevent this — card data goes from the customer browser directly to BoA servers. Do not attempt to intercept or proxy this data.

REQUIRED: Any implementation that routes card data through Crown & Pitch servers will result in immediate merchant account suspension per BoA policy. This applies to the new checkout additions — the existing site infrastructure must not be modified in a way that intercepts payment data.

8.2 Signature & Tamper Prevention

- Sign all request fields except card_number, card_cvn, and signature
- Validate the response signature on every BoA response before acting on it — if signatures do not match, treat the response as malicious and discard it
- Store access keys and secret keys securely — never expose in client-side code or version control
- Security keys expire after 2 years — calendar a renewal reminder
- Do not reuse security keys between test and production profiles

8.3 AVS & CVN

Configure automatic authorization reversals for failed AVS and CVN checks in the Secure Acceptance profile under Payment Settings. This releases held funds on the customer's card when a transaction fails verification.

8.4 Fraud Management

Enable Fraud Management and Verbose Data in the Secure Acceptance profile. Pass customer_ip_address in every transaction request for fraud scoring. Configure device fingerprinting per the BoA Fraud Management documentation available in Business Advantage 360.

9. BoA Secure Acceptance Profile Setup Checklist

Complete all steps in the BoA Business Advantage 360 portal before activating this new checkout feature. Both a TEST profile and a separate PRODUCTION profile are required — profiles cannot be copied between environments.

#	Setup Step
1	Create Secure Acceptance profile — Integration Method: Checkout API
2	Enable Payment Tokenization in profile settings
3	Enable Fraud Management and Verbose Data
4	Add card types: Visa (001), Mastercard (002), Amex (003), Discover (004) — enable USD for each
5	Enable automatic authorization reversal for failed AVS check
6	Enable automatic authorization reversal for failed CVN check
7	Create and download security keys (test and production separately)
8	Configure Merchant POST URL (HTTPS, TLS 1.2+)
9	Configure Email Receipt to Customer with Crown & Pitch sender email and name
10	Configure Send Copy To with Crown & Pitch admin email
11	Configure Customer Response Page URL
12	Activate the profile (Promote Profile button)
13	Complete test transactions using BoA test card numbers before activating this new checkout feature
14	Re-create all profile settings in PRODUCTION environment separately

10. Test Card Numbers

Use the following card numbers in the BoA Demonstration and Certification Environment (DCE) only during development and staging testing. Do not use test card numbers against the live production endpoint. Remove spaces when submitting. Use any future expiry date and any CVN.

Card Type	Test Number
Visa	4111 1111 1111 1111
Mastercard	5555 5555 5555 4444
American Express	3782 8224 6310 005
Discover	6011 1111 1111 1117

11. Pre-Activation Requirements (Brent — Action Items)

The following must be completed by Crown & Pitch before the checkout feature is activated on the live site. These are not developer tasks. The developer should not push the payment feature to production until confirmation is received that all items below are complete.

Action Item	Notes
Notify BoA in writing of intent to charge processing fee	Required 30 days before activating the checkout feature — start this immediately

Confirm exact processing fee label approved by BoA	"Processing Fee" is the current agreed label — get written confirmation
Confirm BoA merchant account is enabled for Secure Acceptance Checkout API	Contact BoA Customer Support if not already enabled
Provide developer with production Profile ID and Access Key	Do not send Secret Key by email — provide through a secure channel
Register for Texas Sales and Use Tax Permit	Free — apply at Texas Comptroller website if not already done
Confirm sales tax rate for each tournament venue location	Use Texas Comptroller Tax Rate Locator — rate may differ from Prosper for away venues

Questions on this brief should be directed to Crown & Pitch LLC before beginning development. The live site must remain fully operational throughout all development and testing phases.

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